



Project Proposal to be Submitted for Independent Project Dissertation
for (Bachelor of Art) Information Technology

Department of Social Statistics

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**“Development of a comprehensive mobile application for export crop
knowledge and trade facilitation”**

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1) Introduction

The global agricultural market is rapidly advancing, and for primary crop-exporting countries, maintaining competitiveness and efficiency has become essential. Sri Lanka, as a country heavily reliant on the export of tea, cinnamon, and coconut, plays a significant role in contributing to the nation's economic growth. However, farmers and exporters engaged in the agricultural export sector face significant challenges in accessing the necessary information.

Currently, existing websites in Sri Lanka, such as those managed by the Sri Lanka Tea Board, Export Development Board, and other institutions, provide basic information. However, this information is not practical or easily accessible to farmers and small-scale exporters. For instance, updates on market prices, favorable trade opportunities, and international trade regulations are not readily available in real-time. Additionally, these websites primarily benefit advanced and high-level users, neglecting the needs of farmers and small-scale users who require timely and relevant information.

This highlights the need for a user-friendly, modern technological solution that provides easy access to crucial information for farmers and exporters in Sri Lanka's agricultural export sector. The lack of an efficient information center leads to delays in export activities, while the absence of up-to-date data poses a risk of missed opportunities.

This project aims to develop a user-friendly mobile application that will provide farmers and exporters with real-time updates on market prices, international trade regulations, and professional advice. Through this app, efficiency and competitiveness in Sri Lanka's agricultural export sector can be enhanced, marking the beginning of a new era in the industry.

2) Literature Review

- 1) The Market Mobile Application developed by Muñoz, Estioco, Zapanta, and Delos Reyes (2020) aims to bridge the gap between local producers and consumers by providing a digital platform for direct market access. This application is particularly beneficial for primary sector workers, such as farmers and fishermen, as it allows them to connect with buyers without the need for intermediaries, thus promoting fairer pricing and more transparent trade practices. The study highlights that by enabling direct engagement, the app not only enhances income stability for rural producers but also fosters economic growth within local communities.

A significant challenge addressed by the Market Mobile Application is the digital divide in rural areas. The lack of reliable internet access and limited digital literacy are common barriers that restrict rural communities from fully benefiting from digital technologies. To counter these challenges, the app incorporates offline capabilities and a user-friendly design, making it accessible even to those with minimal digital skills. This approach aligns with the goals of digital inclusion and empowers local producers to participate actively in the digital economy.

- 2) The integration of mobile phones into the daily activities of small-scale fishermen has proven to be transformative, especially in low-income communities where access to information and resources is limited. According to Saville et al. (2015), mobile phones have become a crucial tool for small-scale fishermen in North Jakarta, Indonesia, enabling them to improve their economic circumstances while enhancing community resilience. These devices serve multiple functions beyond basic communication, such as accessing critical market data, coordinating sales, and connecting with potential buyers, which helps fishermen secure fairer prices and minimize losses from unsold stock. The study highlights that mobile phones allow fishermen to access real-time information about market demand and pricing trends, which enables them to time their sales more effectively and maximize profits.

Beyond the economic benefits, mobile phones are essential for improving safety during fishing activities. Fishermen can use mobile phones to communicate with their peers while at sea, facilitating emergency support and rapid response in hazardous conditions. By enhancing communication within the fishing community, mobile phones foster a

collaborative environment where fishermen can share information about safe fishing zones, weather updates, and potential hazards, further supporting safety and productivity. This use of mobile technology is particularly important in coastal areas of Indonesia, where natural disasters and unpredictable weather patterns are common, and traditional forms of communication are often inadequate.

Moreover, mobile phones also support social and community engagement. Fishermen use mobile phones to maintain social ties, coordinate community activities, and engage in social networks that are critical for their well-being and economic resilience. Saville et al. (2015) emphasize that the use of mobile technology among small-scale fishermen strengthens social bonds and collective action, which are crucial for community support and mutual aid. This integration of mobile technology enhances not only the livelihoods of individual fishermen but also the collective economic stability and social fabric of fishing communities. This study underscores the critical role mobile technology plays in fostering economic empowerment, enhancing safety, and building resilience among small-scale fishermen, especially in regions where traditional communication and information-sharing methods are limited.

- 3) Mobile applications have been transformative tools in supporting agriculture and rural development, especially in developing regions. According to Qiang et al. (2012), mobile applications (m-apps) for ARD offer significant benefits by connecting rural producers directly to market information, extension services, and supply chain management. These applications reduce the asymmetry in information that often disadvantages rural farmers, enabling them to make informed decisions on pricing, crop selection, and market timing. In Kenya, for instance, the application DrumNet enabled farmers to achieve a 32% increase in income through better integration in the supply chain, while Virtual City's AgriManagr enhanced efficiency in tea production and reduced transaction costs by replacing paper records with electronic data entry systems. Such systems also empower rural communities economically by minimizing losses, reducing intermediary exploitation, and enhancing access to financial services and insurance, which are often limited in rural regions.

- 4) The integration of Information and Communication Technology (ICT) in agriculture has been increasingly recognized as a critical tool for enhancing market information access among smallholder farmers, especially in developing regions. According to Mbaya (2023), smallholder farmers often face significant challenges in accessing timely and reliable market information, which is essential for making informed decisions about crop pricing, seasonal demand fluctuations, and quality standards required by various markets. The lack of access to this information places smallholder farmers at a disadvantage, limiting their income potential and making them more vulnerable to exploitation by intermediaries who often control market information channels.

The study on the Viazisoko Digital Platform, which specifically caters to smallholder potato farmers in Buuri Sub County, Meru County, Kenya, illustrates how ICT tools can bridge the gap in market information access. The platform enables farmers to access real-time data on market prices, production tips, and distribution channels. As highlighted by Mbaya (2023), the Viazisoko platform is more than just a pricing tool; it empowers farmers by providing comprehensive crop management guidance, promoting better agricultural practices, and supporting farmers in improving crop quality and yield. By offering direct access to such vital information, ICT tools like Viazisoko help farmers reduce their dependency on middlemen, increase their negotiating power, and make better-informed choices on where and when to sell their produce.

Moreover, ICT tools play a critical role in helping farmers navigate challenges associated with agricultural risks and seasonal price volatility. By accessing data on weather forecasts, soil quality, and pest management through the platform, farmers can make proactive decisions to mitigate potential crop losses. Studies on ICT in agriculture also indicate that these platforms foster a more transparent marketplace, reducing information asymmetry and enabling farmers to participate in a fairer, more competitive market environment. ICT tools therefore represent not only an economic opportunity but also a pathway for social empowerment by promoting a more inclusive agricultural economy.

- 5) In Sri Lanka, the cultivation of potatoes faces numerous challenges, including improper pesticide use, soil fertility degradation, and inefficient post-harvest management. These factors significantly contribute to reduced crop yields and poor-quality produce, ultimately impacting the economic stability of smallholder farmers. To address these issues, Shanadi et al. (2022) developed a mobile-based system to disseminate Good Agricultural Practices

(GAPs) tailored for potato farmers. This initiative integrates technology into agriculture, providing real-time access to critical information, enabling farmers to implement GAPs effectively.

The mobile application focuses on key agricultural areas such as optimal planting techniques, sustainable pest control methods, and fertilizer management. By incorporating GAP guidelines into a digital platform, farmers are empowered to make informed decisions, reducing costs associated with overuse of inputs while improving crop yield and quality. Additionally, eco-friendly practices promoted through the system align with sustainable agriculture goals, addressing both economic and environmental concerns.

The study highlights that adoption of such mobile-based systems not only improves agricultural productivity but also enhances knowledge dissemination in rural farming communities. This technological intervention bridges the gap between traditional practices and modern agricultural advancements, fostering a transition towards data-driven farming. Moreover, the user-friendly design of the mobile application ensures accessibility for farmers with varying levels of digital literacy, contributing to widespread adoption and long-term benefits.

The existing literature demonstrates significant progress in developing mobile applications for smallholder farmers and local markets, but there is a notable lack of focus on export-oriented agricultural sectors. For instance, Shanadi et al. (2022) focus on providing localized farming guidelines but fail to address the unique requirements of export crops, such as international certification compliance and access to global market intelligence. Similarly, Muñoz et al. (2020) emphasize market access within domestic supply chains but do not account for the complexities of export-driven trade. While Bello-Bravo et al. (2015) discuss improving product quality through ICT interventions, the study does not explore sustainability practices or certification processes critical for global trade. Additionally, Qiang et al. (2012) and Meena et al. (2018) highlight the role of mobile applications in bridging information gaps, yet they lack features like predictive analytics, risk management tools, and tailored export-market strategies. This gap highlights the need for a comprehensive mobile application tailored to Sri Lanka's tea, cinnamon, and coconut sectors, integrating global market insights, export certification guidance, and predictive tools to support farmers and exporters in navigating international trade effectively.

3) Research Problem / Problem Statement

- **Research Problem**

Sri Lankan agricultural exporters and local businesses involved in tea, cinnamon, and coconut face difficulties in accessing reliable, centralized information on crop cultivation, export regulations, and market demand. Existing platforms primarily cater to larger-scale businesses, leaving small-scale farmers and exporters without adequate resources. There is a lack of an integrated digital platform that provides easy access to real-time market updates, trade opportunities, and international regulations, limiting their competitiveness and market reach. This research aims to develop a user-friendly mobile application to address these gaps and improve the efficiency of Sri Lanka's agricultural export sector.

- **Problem Statement**

This study seeks to develop a mobile application that serves as both an informational hub and a marketplace for stakeholders in Sri Lanka's tea, cinnamon, and coconut industries. The application will offer detailed data on crop growth conditions, market trends, and export regulations, as well as provide a dedicated platform for buying and selling products derived from these crops. By delivering real-time, centralized information and a commercial channel, the platform aims to improve decision-making, facilitate regulatory compliance, and expand market opportunities for Sri Lankan producers, suppliers, and exporters.

4) Objectives of the Study

The primary objective of this study is to develop a comprehensive mobile application tailored for individuals and businesses involved in the cultivation, production, and export of Sri Lankan crops such as tea, cinnamon, and coconut. This application is designed to act as an all-inclusive resource that offers in-depth information on these crops, including their economic significance, optimal growth conditions, prevailing market trends, and specific regulatory requirements necessary for international trade. By centralizing this information, the application aims to empower producers and exporters with the knowledge they need to enhance crop quality, ensure compliance with export standards, and make informed decisions that contribute to economic growth.

Additionally, the app will integrate a marketplace component, enabling users to directly engage in the buying and selling of products derived from tea, cinnamon, and coconut. This digital marketplace not only connects local producers with potential buyers but also broadens their access to global markets, thereby fostering a more dynamic and competitive agricultural economy. Through this dual focus on information access and marketplace facilitation, the app aims to support sustainable business practices and improve the overall efficiency and profitability of Sri Lanka's agricultural export sector.

Ultimately, this study seeks to address the gaps in accessibility, efficiency, and market connectivity within the agricultural export industry, creating a digital platform that strengthens the global presence and commercial success of Sri Lankan crops. By improving the timeliness and quality of information, and offering direct market access, the app is expected to support small- to large-scale agricultural enterprises, enabling them to navigate and succeed in the competitive international market.

5) Significance of the Study

The significance of this study lies in its potential to revolutionize the agricultural export sector for key Sri Lankan crops - tea, cinnamon, and coconut by leveraging digital technology to enhance market accessibility and decision-making processes for local producers. In a rapidly globalizing economy, the need for centralized, reliable, and easily accessible information on these export crops has become essential. This application serves as a critical resource for producers and exporters, enabling them to access up-to-date information on crop economics, growth requirements, market trends, and compliance with international trade regulations. Such information can lead to improved crop quality, compliance with export standards, and optimized production practices that increase both the profitability and sustainability of Sri Lanka's agricultural sector.

Furthermore, the application's built-in marketplace function directly supports economic growth by connecting local producers and exporters with global buyers, creating new market opportunities and enhancing competitiveness on an international scale. By offering a platform where stakeholders can buy and sell products derived from these crops, the study addresses barriers in market connectivity and provides a solution that fosters direct and efficient trade.

This study also holds significance for the broader agricultural community by promoting technology adoption in traditional sectors, paving the way for a digital transformation that can increase efficiency and transparency. By demonstrating the effectiveness of a dedicated mobile application in improving both market access and information availability, this study can serve as a model for future innovations in other sectors and regions, ultimately contributing to sustainable agricultural development and economic resilience in Sri Lanka.

6) Methodology of the Study (Study Design & Setting)

This study will employ a mixed-methods design, integrating both qualitative and quantitative research approaches to gather comprehensive insights into the development, usability, and effectiveness of a mobile application for promoting and managing the export of crops such as tea, cinnamon, and coconut. The methodology will be structured in three key phases:

1. Needs Assessment and Market Research Phase (Qualitative and Quantitative Analysis)

- **Qualitative Analysis:** The first step involves conducting semi-structured interviews with stakeholders in the crop export industry, including exporters, agricultural experts, and farmers. The aim is to identify their specific challenges, requirements, and expectations regarding the export process. Additionally, a review of existing systems and platforms will be carried out to determine gaps that the new mobile application can address.
- **Quantitative Analysis:** Surveys will be distributed to a broader group of stakeholders, including small and medium-scale exporters. These surveys will gather data on current practices, usage of technology in the trade process, and specific features needed in the mobile application to improve their operational efficiency.

2. App Design and Development Phase (System Design and Prototyping)

- **Prototype Development:** Based on insights from the needs assessment and market research, an initial prototype of the mobile application will be developed. The prototype will focus on providing features that address the key needs of users, such as market trends, export regulations, crop statistics, and real-time pricing data.
- **Agile Development Methodology:** The application will be developed using an Agile methodology to ensure iterative improvements based on feedback. A small group of users will be involved in testing the prototype in real-world scenarios, providing feedback on usability and effectiveness.
- **User-Centered Design:** The development process will prioritize the needs of the users, focusing on simplicity and accessibility. The design will include a user-friendly

interface with easy navigation, and features like real-time updates and notifications for market trends.

3. Usability Testing and Evaluation Phase

- **Usability Testing:** The app will undergo thorough usability testing with a sample group of stakeholders who are engaged in the crop export process. Usability testing will focus on ease of use, navigation, and user experience. This phase will also assess how well the app meets the needs of different users (e.g., farmers, exporters).
- **Data Collection:** Data will be collected using both qualitative methods (interviews) and quantitative methods (surveys and app usage analytics).
- **Analysis:** Usability data will be analyzed to identify patterns, issues, and areas for improvement. Metrics such as user satisfaction, task completion rates, and feedback on user experience will be evaluated.

Study Setting

The study will take place in the context of Sri Lanka's agricultural export sector, specifically focusing on regions where tea, cinnamon, and coconut farming are prevalent. The mobile application will be tested in both rural and urban areas to understand the diverse needs of users, from small-scale farmers to large exporters. The platform will be designed to cater to the specific needs of Sri Lanka's agricultural industry while also being adaptable to other markets as needed.

7) Problems, Limitations and Delimitations

One key issue is the lack of centralized information on tea, cinnamon, and coconut exports, particularly for small-scale farmers and exporters in rural areas with limited access to updated guidelines and market trends. Digital literacy may also impact adoption, as some users might struggle with platform navigation. Additionally, data accuracy and frequency of updates are essential for timely decision-making, yet frequent updates may require significant resources. Internet connectivity in rural areas remains a constraint, affecting platform reliability. The dual-language support in Sinhala and English promotes inclusivity but could face challenges in fully localizing technical terms.

Limitations in the project's scope include a focus on Sri Lankan export information, with limited access to real-time global data. Furthermore, the app relies on public data sources and does not include live trading insights or recent regulatory changes behind paywalls.

The project is delimited to tea, cinnamon, and coconut exports, targeting small and medium-sized exporters. Initially, language support is restricted to Sinhala, with possible expansions as needed. Information is based on publicly available, credible sources to provide reliable, accessible content to Sri Lankan exporters. This approach aims to establish an accessible, reliable resource foundation for the local export industry.

8) Time Frame / Time Span

Activity/Task	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
1. Select a topic									
2. Finalizing research proposal									
3. Literature Review									
4. Designing Research instrument									
5. Data Collection									
6. Data Analysis									
7. Review and Revision									
8. Final Submission									

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